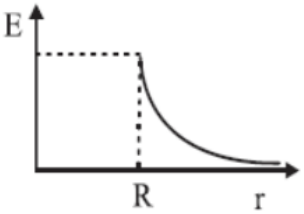
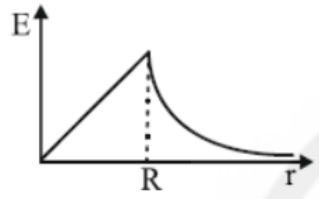
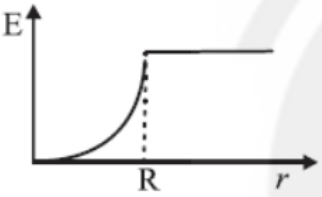
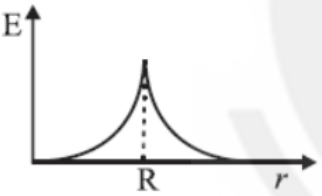
Q1 Which of the following graphs shows the variation of magnitude of electric field E due to a hollow spherical conductor of radius R as a function of distance from the center of the sphere?

(A) 

(B) 

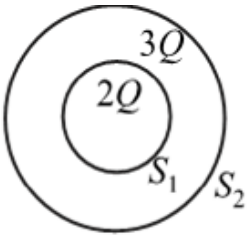
(C) 

(D) 

Q2 When 20 μC charge is placed inside a closed surface then electric flux linked with the surface is ϕ. If 40 μC charge is now put inside the closed surface containing 20 μC charge, then new electric flux linked with the surface is:

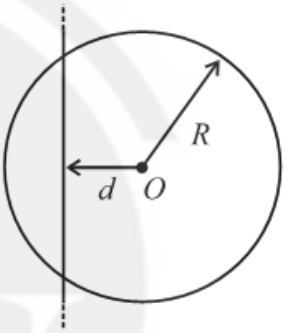
(A) 4ϕ (B) 5ϕ
(C) 3ϕ (D) 8ϕ

Q3 S_1 and S_2 are two concentric conducting shells having charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of the electric flux through S_1 and S_2 is:



(A) 2 : 3 (B) 2 : 5
(C) 3 : 2 (D) 5 : 2

Q4 Calculate the electric flux through the Gaussian spherical surface of radius R, when an infinitely long line charge of linear charge density (λ), lies on it, at a distance d (< R) from its centre:



(A) $\frac{\lambda\sqrt{R^2-2d^2}}{\epsilon_0}$
(B) $\frac{2\lambda\sqrt{R^2-d^2}}{\epsilon_0}$
(C) $\frac{\lambda\sqrt{R^2-4d^2}}{\epsilon_0}$
(D) $\frac{2\lambda\sqrt{R^2+d^2}}{\epsilon_0}$

Q5 The electric intensity outside a charged sphere of radius R at a distance $r(r > R)$ is

(A) $\frac{\sigma R^2}{\varepsilon_0 r^2}$
(B) $\frac{\sigma r^2}{\varepsilon_0 R^2}$
(C) $\frac{\sigma r}{\varepsilon_0 R}$
(D) $\frac{\sigma R}{\varepsilon_0 r}$

Q6 A hollow metal sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre

(A) Increases as r increases for $r < R$ and $r > R$ for

(B) Zero as r increases for $r < R$, decreases for r increases as $r > R$

(C) Zero as r increases for $r < R$, increases for r increases as $r > R$

(D) Decreases as r increases for $r < R$ and $r > R$ for

Q7 The electric field in a certain region is acting radially outward and is given by $\mathbf{E} = \mathbf{Ar}$. A charge contained in a sphere of radius ' a ' centered at the origin of the field, will be given by

(A) $\mathbf{A}\varepsilon_0\mathbf{a}^2$

(B) $4\pi\varepsilon_0\mathbf{Aa}^3$

(C) $\varepsilon_0\mathbf{Aa}^3$

(D) $4\pi\varepsilon_0\mathbf{Aa}^2$

Q8 Two infinitely long parallel conducting plates having surface charge densities $+\sigma$ and $-\sigma$, respectively, are separated by a small distance. The medium between the plates is vacuum. If ε_0 is the dielectric permittivity of vacuum, then the electric field in the region between the plates is

(A) $\mathbf{0}$ volts/rmetre

(B) $\frac{\sigma}{2\varepsilon_o}$ volts/metre

(C) $\frac{\sigma}{\varepsilon_o}$ volts/metre

(D) $\frac{2\sigma}{\varepsilon_o}$ volts/metre

Q9 The electric field near a conducting surface having a uniform surface charge density is given by

(A) $\frac{\sigma}{\varepsilon_0}$ and is parallel to the surface

(B) $\frac{2\sigma}{\varepsilon_0}$ and is parallel to the surface

(C) $\frac{\sigma}{\varepsilon_0}$ and is normal to the surface

(D) $\frac{2\sigma}{\varepsilon_0}$ and is normal to the surface

Q10 A uniformly charged conducting sphere of $4.4\ \mathbf{m}$ diameter has a surface charge density of $60\mu\mathbf{Cm}^{-2}$. The charge on the sphere is

(A) $7.3 \times 10^{-3}\mathbf{C}$

(B) $3.7 \times 10^{-6}\mathbf{C}$

(C) $7.3 \times 10^{-6}C$

(D) $3.7 \times 10^{-3}C$

Answer Key

Q1	A
Q2	C
Q3	B
Q4	B
Q5	A
Q6	B
Q7	B
Q8	C
Q9	C
Q10	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

(1)

We know that Electric field for hollow spherical conductor is given by

$E = 0$, for $r < R$

$E = \frac{Q}{4\pi\epsilon_0 r^2}$ for $r \geq R$

Hence,

we can say that for a conductor Electric filed will be zero inside and outside it will be varying to $E\ \alpha\ \frac{1}{r^2}$.

Video Solution:

Q2 Text Solution:

(3)

$\phi = \frac{20\times10^{-6}}{\varepsilon_0}$,

$\phi' = \frac{(20+40)\times10^{-6}}{\varepsilon_0}$

$= \frac{3\times20\times10^{-6}}{\varepsilon_0} = 3\phi$

Video Solution:

Q3 Text Solution:

(B)

Flux through sphere $S_1, \phi_1 = \frac{\text{charge}}{\epsilon_0} = \frac{2Q}{\epsilon_0}$

and

flux through sphere $S_2, \phi_2 = \frac{2Q+3Q}{\epsilon_0} = \frac{5Q}{\epsilon_0}$

$\therefore$ Ratio $\left(\frac{\phi_1}{\phi_2}\right) = \frac{\frac{2Q}{\epsilon_0}}{\frac{2Q}{\epsilon_0}} = \frac{2}{5}$

[New NCERT Class 11th Page No. 21, 22 and 23]

Video Solution:

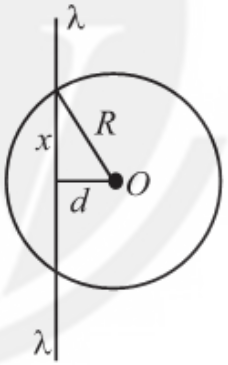
Q4 Text Solution:

(2)

$\frac{\phi=2\lambda\sqrt{R^2-d^2}}{\epsilon_0}$

OR

$\phi = \frac{\Sigma q}{\epsilon_0}$



Σq =change enclosed by Gaussian surface.

$\Sigma q = \lambda$(length of wire enclosed)

$= \lambda\left(2\sqrt{R^2-d^2}\right) \Rightarrow \phi = \frac{2\lambda\sqrt{R^2-d^2}}{\epsilon_0}$

Video Solution:

Q5 Text Solution:

For $r > R$ a uniformly charged sphere acts as a point charge Q at its centre.

$$Q = \sigma \times 4\pi R^2$$

$$E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\sigma 4\pi R^2}{r^2} = \frac{\sigma R^2}{\varepsilon_0 r^2}.$$

Video Solution:



Q6 **Text Solution:**

For a uniformly charged conducting shell (radius R):

- **Inside the metal cavity ($r < R$)**
  Charge resides on the outer surface
  enclosed charge = 0
  $E = 0$ (independent of r).
- **Outside the shell ($r > R$)**
  Field is that of a point charge Q at the centre:
  $E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$, which **decreases** as r increases.

Hence E is zero for all $r < R$ and falls off with r for $r > R$.

Video Solution:



Q7 **Text Solution:**

Gauss's law: $q = \varepsilon_0\,\Phi$
For a sphere of radius a

$$E = Ar \implies E(a) = Aa$$

Flux through the spherical surface

$$\Phi = E(a)\,(4\pi a^2) = Aa\,(4\pi a^2) = 4\pi A a^3$$

Enclosed charge

$$q = \varepsilon_0 (4\pi A a^3) = 4\pi\varepsilon_0 A a^3$$

Video Solution:



Q8 **Text Solution:**

Field of a single infinite sheet:
$E = \sigma/(2\varepsilon_0)$.
Between the two plates the fields from $+\sigma$ and $-\sigma$ point the same way, so

$$E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \boxed{\frac{\sigma}{\varepsilon_0}}\ \mathrm{V\,m^{-1}}.$$

Video Solution:



Q9 **Text Solution:**

Thought for a couple of seconds
For a small Gaussian pill box straddling a charged conducting surface:

$$\oint \mathbf{E}\cdot d\mathbf{A} = E(\Delta A) = \frac{\sigma\,\Delta A}{\varepsilon_0} \implies E = \frac{\sigma}{\varepsilon_0},$$

directed perpendicular to (normal to) the surface; inside the conductor $E = 0$.

Option $\frac{\sigma}{\varepsilon_0}$, normal to the surface

Video Solution:

Q10 **Text Solution:**

Surface charge density

$\sigma = 60\ \mu\text{C m}^{-2} = 60 \times 10^{-6}\ \text{C m}^{-2}$

Radius $r = \frac{4.4\ \text{m}}{2} = 2.2\ \text{m}$

Surface area

$A = 4\pi r^2 = 4\pi(2.2)^2 = 4\pi(4.84)$

$= 19.36\pi\ \text{m}^2$

Charge

$Q = \sigma A = (60 \times 10^{-6})(19.36\pi) \approx 3.7 \times 10^{-3}\ \text{C}.$

Video Solution:

Android App | iOS App | PW Website